

help organisations to improve the agility, scalability, reliability, and security of their applications.

Containers and serverless cloud technologies both offer benefits for developing, deploying, and managing applications. While there are pros and cons to each, the decision of which to use ultimately comes down to the specific use case and requirements. Use containers when you have a complex application with multiple services that require a consistent and isolated environment. Containers are also ideal for testing and development environments. Use serverless when you have an event-driven application or service that requires encapsulation and scalability. Serverless is also ideal for applications with unpredictable usage patterns or a need for ultra-fast response times. By carefully considering your options and taking advantage of the

strengths of each technology, you can build resilient, scalable, and efficient

applications that meet the needs of your users and business goals. **END** 

References

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Disclaimer

This article expresses the view of the author and not of the organisation he works in.

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5. Macie

- Scans the S3 buckets for sensitive information (personally identifiable information, etc)

6. Inspector

- Performs a vulnerability analysis at scale for software vulnerabilities and network exposure

7. Guard Duty

- Uses machine learning techniques to identify suspicious user and resource behaviour

8. Detective

- Can analyse logs for security issues

9. Firewall Manager

- Enables centralised management of firewall rules across various accounts in AWS organisations

- Virtual server monitoring
 - Virtual network monitoring
 - Container monitoring (Docker/ Kubernetes, etc)
 - Database monitoring
 - Cloud activity monitoring
- The key characteristics of an IaaS cloud monitoring system are:
- Should be highly available to keep track of other systems and services
 - Provide real-time notifications and alerts on outages and service degradation
 - Immediately detect service outages and unauthorised activities
 - Should be highly scalable to meet the demands of spikes in activity
 - Be able to handle and process huge amounts of monitoring data
 - Be able to collect standardised metrics using vendor-neutral standards
 - Should support native and hybrid

cloud services and applications

- Should allow end-to-end traceability
 - Should have an intuitive UI for usability
 - Should be able to integrate with visualisation tools
- Prometheus, Icinga and Cacti are some of the most popular open source tools used for monitoring IaaS services.

IaaS security and monitoring is an emerging field, and cyber security is a niche skill. The nature (and scope) of cyber threats is on the rise. Organisations should follow the security-by-design approach, and make security and monitoring an intrinsic part of their develop-build-test-deploy activity. When properly architected and designed by including the security principles, organisations can build robust applications. **END** 

Monitoring IaaS

Monitoring, management and governance are key elements of IaaS cloud infrastructure. This includes, but is not limited to:

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